Peihao Zhu

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Work Experience

Foundation Model Team, Adobe — San Jose, CA
Senior Applied Scientist — *Jan 2026 – Present*

Seed Team, ByteDance — San Jose, CA
Senior Research Scientist — *Sep 2024 – Jan 2026*

Seed Team, ByteDance — San Jose, CA
Research Scientist — *Jul 2023 – Aug 2024*

Reality Lab, Meta — Burlingame, CA
Research Scientist Intern — *Oct 2022 – Feb 2023*

Creative Vision Lab, Snap — Los Angeles, CA
Research Scientist Intern — *May 2022 – Sep 2022*

Education

King Abdullah University of Science and Technology (KAUST) — Kingdom of Saudi Arabia
PhD in Computer Science — *Dec 2019 – Jun 2023*

King Abdullah University of Science and Technology (KAUST) — Kingdom of Saudi Arabia
Master of Science in Computer Science — *Aug 2017 – Dec 2019*

Institute of Automation, Chinese Academy of Sciences — China
Master's Candidate in Computer Science — *Sep 2016 – Jul 2017*

Northeastern University — China
Bachelor of Technology in Automation — *Aug 2012 – Jun 2016*

Projects

Adobe Video Foundation Model – Gen6 / Gen6-long [Website]
Core Contributor, Video Foundation Models — *Jan 2026 – Present*

- Serve as a main contributor to Adobe's next-generation video foundation model, leading the development and implementation of Gen6-long from research exploration to model delivery.
- Drive improvements in long-video visual and temporal consistency while optimizing model training and inference for greater computational efficiency.

MMCORE – Multimodal Image Editing Foundation Model [Paper] [Website]
Core Contributor, Multimodal Image Editing — *Jul 2025 – Dec 2025*

- Drove core model development and large-scale experimentation for MMCORE, a unified multimodal foundation model for high-quality image editing.
- Contributed across model training, evaluation, and research iteration, helping advance multimodal understanding and instruction-guided editing.

Seedance 1.0 – Video Generation Foundation Model [Paper] [Video] [Website]
Core Contributor, Video Foundation Models — *Dec 2024 – Jun 2025*

- Served as a main contributor to the development and large-scale training of Seedance 1.0, helping build the data and representation infrastructure underlying the video foundation model.
- Designed and implemented the distributed data-loading system and large-scale VAE-latent and text-embedding precomputation pipelines, enabling efficient preprocessing, storage, and consumption of training features.

Seaweed-7B – Video Generation Foundation Model [Paper] [Video] [Website]
Founding/Core Contributor, Video Foundation Models — *Mar 2024 – Nov 2024*

- Built Seaweed-7B from its early stage as a main contributor, driving key components of the model and large-scale training stack through research iteration and public release.
- Designed the VAE for efficient visual-information compression and built the distributed data-loading system for reliable, high-throughput video-model training.

TikTok AI-moji [Blog] [Platform]
Core Contributor, Generative Avatars — *Jul 2023 – Dec 2024*

- Drove generative-avatar model development for TikTok AI-moji, contributing from research prototyping through product deployment.
- Developed personalization and generation capabilities for creating stylized avatars from user images at product scale.

PUBLICATIONS

MoviEdit: Long-form Multi-Shot Video Editing Made Easy
Suhyeon Lee, Zhengang Li, Soo Ye Kim, Peihao Zhu, et al., Hyeonho Jeong
ArXiv pre-print, 2026 [Paper] [Webpage]

MMCORE: MultiModal COnnection with Representation Aligned Latent Embeddings
ByteDance Seed | Core Contributor
Technical Report, 2026 [Paper] [Webpage]

Seedance 1.0: Exploring the Boundaries of Video Generation Models
ByteDance Seed | Core Contributor
Technical Report, 2025 [Paper] [Webpage]

Seaweed-7B: Cost-Effective Training of Video Generation Foundation Model
ByteDance Seed | Core Contributor
Technical Report, 2025 [Paper] [Webpage]

Towards Robust Sequential Decomposition for Complex Image Editing
Zilai Zeng, Mingdeng Cao, Zijie Li, Xiaochen Lian, Yichun Shi, Peihao Zhu, Chen Sun, Peng Wang
Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2026 [Paper] [Webpage]

3DAvatarGAN: Bridging Domains for Personalized Editable Avatars
Rameen Abdal, Hsin-Ying Lee, Peihao Zhu, Menglei Chai, Aliaksandr Siarohin, Peter Wonka, Sergey Tulyakov
Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2023 [Paper] [Webpage]

HairNet: Hairstyle Transfer with Pose Changes
Peihao Zhu, Rameen Abdal, John Femiani, Peter Wonka
Proc. European Conference on Computer Vision (ECCV), 2022 [Paper] [Webpage]

CLIP2StyleGAN: Unsupervised Extraction of StyleGAN Edit Directions
Rameen Abdal, Peihao Zhu, John Femiani, Niloy Mitra, Peter Wonka
SIGGRAPH Conference Proceedings, 2022 [Paper] [Webpage]

Mind the Gap: Domain Gap Control for Single Shot Domain Adaptation for Generative Adversarial Networks
Peihao Zhu, Rameen Abdal, John Femiani, Peter Wonka
International Conference on Learning Representations (ICLR), 2022 [Paper] [Webpage]

Barbershop: GAN-based Image Compositing using Segmentation Masks
Peihao Zhu, Rameen Abdal, John Femiani, Peter Wonka
ACM Transactions on Graphics (Proc. SIGGRAPH Asia), 2021 [Paper] [Webpage]

Improved StyleGAN Embedding: Where are the Good Latents?
Peihao Zhu, Rameen Abdal, Yipeng Qin, John Femiani, Peter Wonka
ArXiv pre-print, 2021 [Paper] [Webpage]

Styleflow: Attribute-conditioned exploration of stylegan-generated images using conditional continuous normalizing flows
Rameen Abdal, Peihao Zhu, Niloy Mitra, Peter Wonka
ACM Transactions on Graphics (TOG), 2021 [Paper] [Webpage]

Labels4Free: Unsupervised Segmentation using StyleGAN
Rameen Abdal, Peihao Zhu, Niloy J. Mitra, Peter Wonka
Proc. IEEE International Conference on Computer Vision (ICCV), 2021 [Paper] [Webpage]

Flow-Guided Video Inpainting with Scene Templates
Dong Lao, Peihao Zhu, Peter Wonka, Ganesh Sundaramoorthi
Proc. IEEE International Conference on Computer Vision (ICCV), 2021 [Paper] [Webpage]

SEAN: Image Synthesis with Semantic Region-Adaptive Normalization
Peihao Zhu, Rameen Abdal, Yipeng Qin, Peter Wonka
*Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR **Oral**), 2020* [Paper] [Webpage]

Large Scale Architecture Asset Extraction from Panoramic Imagery
Peihao Zhu, Wamiq Reyaz Para, Anna Fruehstueck, John Femiani, Peter Wonka
IEEE Transactions on Visualization and Computer Graphics (TVCG), 2020 [Paper] [Webpage]

Awards & Honors

- **KAUST CEMSE Dean's List Award** *2021-2022*
- **National Scholarship** *2013*
- KAUST Fellowship *2017*
- UCAS Scholarship *2016*
- Scholarship for Outstanding Merits *2012-2016*

Reference

Prof. Peter Wonka
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http://peterwonka.net